INCH-POUND

MIL-C-71155 (AR)
AMENDMENT 3
31 July 1996
SUPERSEDING
AMENDMENT 2
05 December 1995

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M987
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-71155 (AR), dated 01 November 1993, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Delete: "INSPECTION EQUIPMENT DRAWINGS

9278622 - Envelope Drawing for Leak Testing of Ignition Cartridges".

PAGE 3

3.4.1, line 2: Delete: "8.85" and substitute: "1.00".

3.4.2, at end of sentence: Delete: "4.5.2.2" and substitute: "4.5.2.1".

3.5: Delete in its entirety.

PAGE 7

TABLE I: Delete: "Static pressure (b) ... 25 ... 3.8 ... 4.5.7".

AMSC N/A **FSC 1315**

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PAGE 8

TABLE I, under Shell, Shotgun, Primed, 12 Gauge Assy.:

Delete:

"Sensitivity test ... 20 (a) ... 3.4.1/3.4.2 ... 4.5.2"

and substitute:

"Sensitivity test ... 700 (a) ... 3.4.1/3.4.2 ... 4.5.2".

Delete:

"Cartridge, Ignition, M986 (prior to addition of powder)"

and substitute:

"Cartridge, Ignition, M987 (prior to addition of powder)".

Delete "Leak test ... 50 (b) ... 3.5 ... 4.5.3".

On the line "Velocity, functioning and security": Delete:

"4.5.3" and substitute: "4.5.5".

NOTES: Delete Note (a) in its entirety and substitute:

"(a) Primers for the first article and be from the intended first production lot and thereby negating the necessity of testing the primers from that production lot."

PAGE 12

4.4.2.1, Critical 1: Delete in its entirety.

Add Major 105 as follows:

"105 Sensitivity non-function ... (a) ... 3.4.1 ... Test/4.5.2".

PAGE 14

4.4.2.3, Major 104: Delete in its entirety.

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4.4.3.2.1: Delete in its entirety and substitute:

"4.4.3.2.1 Sensitivity test. Beginning with the first article (primers for the first article can be from the intended first production lot, thereby negating the necessity of testing the primers from the first production lot), and from the first production lot produced, and continuing until three (3) consecutive lots have complied with 3.4.1 and 3.4.2, a sample of 700 primers shall be selected and be subjected to this test. If the sum of the average critical height (H average) and five (5) times the standard deviation(s) multiplied by the weight of the ball exceeds the maximum requirement, or if the sum of the average critical height (H average) less two (2) times the standard deviation(s) multiplied by the weight of the ball exceeds the minimum requirement, the inspection lot shall be rejected."

4.4.3.2.1.1: Delete in its entirety.

PAGE 17

4.4.3.2.2: Delete in its entirety.

4.4.3.2.2.1: Delete in its entirety.

PAGE 18

4.4.3.5: Delete in its entirety.

PAGE 19

4.5.2.1: Delete in its entirety and substitute:

"4.5.2.1 Sensitivity test. The sample of 700 primers shall be divided into 14 groups of 50 primers each and tested in the following manner:

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The first group (50) of primers shall be tested by dropping a steel ball 15/16 inch in diameter and weighing 1.94 ± .02 ounces on the firing pin of the test fixture from a height of nine (9) inches. (The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball.) The primers that fire, misfire, hangfire, or squib shall be recorded. This procedure shall be repeated at successively increasing heights of one (1) inch, using a new group of 50 primers at each different height, until a height is reached at which all primers in the group being tested fire. The test fixture shall then be set to a height of eight (8) inches and the above procedure repeated at successively decreasing heights of one (1) inch, using a new group of 50 primers at each different height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection of the lot shall be determined from calculations based on recorded data. The recorded test data shall be used in the calculation of "H average" and "s" as follows:

H average = A plus B plus C
s = I times the square root of L minus M

where:

A = the sum of Column 3.
B = the first drop height at which all elements misfired.
C = one-half the drop height decrement (absolute value).
M = the square of the sum of Column 3.
I = the drop height decrement (absolute value).

4.5.2.1.1: Tabulation of data. Divide a sheet of paper into columns and tabulate the data as follows:

a. Title Column 1 "Height of Drop" and list all drop heights in consecutive order starting with the lowest height. DO NOT list the heights at which all elements fired or misfired.

b. Title Column 2 "Number of Failures" and list the sum of the number of elements that misfired, squibbed or hangfired at each intermediate height.

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c. Title Column 3 "Decimal Fraction of Elements Failing to Fire at Each Intermediate Height" and record the decimal fraction that failed to fire at each intermediate height. This number is obtained by dividing the value obtained under "Number of Failures" by the number of elements tested at that height. Calculations shall be carried out and recorded to the nearest one-hundredth (.01).

d. Title Column 4 "Standard Deviation Factor" and list odd numerals in sequence, starting with 1, for as many drop heights that are listed in Column 1.

e. Title Column 5 "Standard Deviation Factor Times Fraction of Failures" and, by multiplying each decimal fraction in Column 3 by the number on the same line in Column 4, enter the product on the corresponding line in Column 5.

4.5.2.1.2 Procedures for solving equations. The procedures for solving the equations are as follows:

a. To obtain "A": Add the values in Column 3 and substitute the sum for "A".

b. To obtain "B": The first drop height at which all elements failed to fire and substitute the product for "B".

c. To obtain "C": Divide the drop height increment by 2 and substitute the dividend for "C".

d. To obtain "L": Add the values in Column 5 and substitute the sum for "L".

e. To obtain "M": Square the values in Column 3 and substitute the sum for "M".

f. To obtain "H average": Add the values substituted for A, B, and C.

g. To obtain "s": Subtract the value substituted for "M" from the value substituted for "L" and the obtain the square root of the difference and then multiply the square root by the height of the increment.

h. Using the calculated values for "H average" and "s", if "H average" plus 5 "s" multiplied by the maximum weight of the ball is equal to or less than the maximum inch ounce energy required, and "H average" minus 2 "s" multiplied by the minimum weight of the ball is equal to or greater than the minimum energy value required, the lot is considered acceptable for sensitivity.

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4.5.2.1.3 Definitions. The following definitions apply to the non-function test.

a. Inch ounces: The drop height in inches multiplied by the weight of the ball in ounces.

b. Failure: The following modes apply:

(1) Misfire: A failure wherein initiation of the primer mixture does not occur.

(2) Squib: A failure wherein a burning of the primer mixture occurs without detonation.

(3) Hangfire: A failure wherein an audible delay occurs between the instant application of energy and the detonation report."

4.5.2.2: Delete in its entirety.

4.5.3: Delete in its entirety.

PAGE 20

4.5.5 a.: Delete the last sentence and substitute:

"The cartridges shall be fired at charge 3."

* 4.5.5, paragraph b. Delete the last line in its entirety.

PAGE 21

* 4.5.6.2: Delete the last line in its entirety and substitute:

"The means and standard deviations of the velocities at all firing temperatures shall be calculated using corrected data."

PAGE 22

4.5.7: Delete in its entirety.

PAGE 25

Figure 1: Delete in its entirety.

PAGE 26

Figure 2: Delete in its entirety.

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